

Expression of Interest MARVEL End User Application & Utilization

1. INTRODUCTION

Battelle Energy Alliance, LLC (BEA) Management & Operating Contractor of the U.S. Department of Energy (DOE) owned Idaho National Laboratory (INL), is seeking an Expression of Interest (EOI) from industry providers to suggest potential tests and experiments leveraging the Microreactor Application Research Validation and Evaluation (MARVEL) demonstration. This will consist of a two-step process starting with a feasibility assessment, then a physical implementation, subject to Congressional Appropriation.

2. BACKGROUND

The MARVEL will be an operational microreactor and is expected to serve as a nuclear test bed to demonstrate microreactor operations and end-use applications, paving the way for unleashing private sector nuclear energy development. A primary goal of the project is to provide a platform that can be leveraged by the private sector to mature novel nuclear technologies and demonstrate innovative new use cases for microreactors. Additional information on MARVEL's utilization plan can be found in: <https://www.osti.gov/biblio/2371820>. In short, the MARVEL reactor can be used to the following non-exhaustive list of potential applications:

1. Demonstrating novel nuclear-generated electricity applications (e.g., nuclear-powered microgrids, or nuclear-powered data-centers)
2. Demonstrate novel nuclear heat applications (e.g., chemical processing, synfuel production, novel power conversion systems)
3. Requesting data access (e.g., analyses conducted, documentations, engineering drawings)
4. Operating the reactor in specific modes to validate models (e.g., conducting a specific transient to benchmark codes)
5. Deploy advanced instrumentation and control paradigm to nuclear reactors (e.g., semi-autonomous operations, far-field remote operations)
6. Demonstrate novel safeguards and security paradigms for nuclear technology (e.g., cyber-secure communications, novel safeguard techniques for microreactors)

The current MARVEL scope will consist of developing an 85kWth reactor producing more than 10 kWe of electricity in the first stage of operations. In the second stage, process heat (above 300°C) will be provided to end-users specifically for type 2 demonstrations from the list above. Additional information on the technical specifications can be found in the MARVEL factsheets.

3. SUBMITTAL REQUIREMENTS

INL is soliciting requests from private industry on potential tests or demonstrations, leveraging MARVEL, that would be valuable for them to conduct. Companies or other interested parties are encouraged to submit EOIs relating to any of the above-mentioned topical areas (or other topics not covered by the list). Expressions of Interest will be ranked based on their potential impact to the broader US energy landscape. Promising concepts will be invited to respond to a potential Notice of Opportunity (NO) at a later stage. The NO will contain additional information about requirements for a more detailed proposal, funding expectations, timeline, and proposal ranking metrics. Following selection, a two-step process is envisaged whereby stakeholders work with national lab staff first to ensure the viability of the test/demonstration (currently intended for Fiscal Year 2026). This is then followed by a solicitation to conduct physical test/demonstration (currently intended for Fiscal Year 2027-2028).

Interested entities are requested to provide expressions of interest (via email) by close-of business on 08/30/2025. Responses must include sufficient written information confirming the responder's qualifications and the relevancy of their proposed scope based on the guidance in Section 2. Note that responses must be no more than 2 pages in length in PDF format. They should clearly state which of the above 6 category they fall under, which partnering institutions are included in the proposal, and what level of support (beyond the NO) is needed from INL (e.g., if the end-user or INL is expected to provide specific hardware). The content should ideally cover: the goal of the application, why it is impactful for the US nuclear industry, potential technical needs and technical specifications (e.g., quantity of energy needed, temperature range, MARVEL reactor operating conditions, test durations).

If you have questions, or need additional information, please contact Stacie Strain, Program Manager, via email at stacie.strain@inl.gov with the subject line "Response to MARVEL Utilization EOI." Technical questions must be submitted in writing via email. Only potentially qualified entities, as determined by BEA, will be contacted for further discussions or the release of a request for proposal for the project.

4. PROPRIETARY INFORMATION

Because information received in response to this EOI may be business sensitive or proprietary to the responding party, respondents are encouraged to minimize the inclusion of sensitive information to the extent possible and mark information in their responses accordingly.